Joseph E. Velasquez

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Education

California State University Fullerton — **Fullerton, CA**
B.S. Computer Science — *Aug 2025*
Riverside City College — **Riverside, CA**
A.S. Computer Science — *May 2022*

Experience

Inland Neurosurgery Institute - Software and Systems Engineer
Aug 2025 – Oct 2025

- Supported integration and deployment of a new Electronic Medical Records (EMR) system, ensuring minimal disruption to clinical operations
- Resolved hardware, software, and network issues during EMR migration to ensure continuity of clinical operations
- Built and maintained automation workflows using Microsoft Power Automate, reducing manual administrative workload and increasing efficiency
- Assisted in maintaining cybersecurity measures, including system monitoring, patching, and basic threat mitigation
- Developed a patient scheduling chatbot using local LLM, enabling automated appointment booking and reducing administrative workload

Residential IT Support Technician (ResNet) - University of California, Santa Barbara (UCSB)
Aug 2018 – June 2019

- Provided Tier 1 technical support for hardware, software, and network connectivity issues in a residential campus environment.
- Diagnosed and resolved Windows, macOS, Wi-Fi, Ethernet, TCP/IP, DNS, and DHCP issues.

Projects

Homelab Cloud + AI Platform - Self-Hosted Kubernetes/GitOps Infrastructure
Sep 2026

- Deployed a two-node Kubernetes (K3s) cluster with GitOps via Argo CD across a Raspberry Pi 5 and GPU desktop, with CI/CD (GitHub Actions, multi-arch image builds) and secrets encrypted at rest with SOPS
- Found and fixed a production-style security gap where firewall rules never applied to Kubernetes traffic due to iptables chain ordering, verified with real blocked-vs-allowed traffic tests
- Built a FastAPI backend (rate limiting, retries, Prometheus metrics) with a RAG pipeline over a fully local Ollama LLM, backed by PostgreSQL/pgvector and Redis
- Load tested with k6 (0% errors at ~480 req/s sustained) and ran real failure injection (pod kills, dependency outages, a stopped AI backend), verifying results against Prometheus/Grafana rather than client output alone
- Built and shipped a full-stack cluster status dashboard (FastAPI + vanilla JS) with its own pytest suite and GitHub Actions CI/CD pipeline (test, multi-arch container build, auto-deploy via GitOps), integrating live Prometheus metrics and SSH-triggered remote infrastructure control

ParkingPulse - AI Vehicle Tracking & Data System
Aug 2024 – May 2025

- Engineered a real-time AI vehicle tracking platform using Python, OpenCV, and YOLOv8 for intelligent object detection and analytics.
- Designed and implemented RESTful APIs to handle real-time data ingestion, processing, and retrieval
- Integrated Firebase for authentication and cloud-based data storage, enabling scalable and secure data management
- Optimized system performance by reducing processing latency through frame skipping and model tuning
- Implemented data pipelines for continuous processing and storage of vehicle detection events

RAG Office Assistant - Local-LLM Retrieval-Augmented Chatbot
Sep 2025

- Built a retrieval-augmented generation (RAG) chatbot in FastAPI, using FAISS vector search and sentence-transformer embeddings to ground responses in a real knowledge base instead of relying on model hallucination
- Integrated Ollama for fully local LLM inference so patient data and queries never leave the machine, streaming responses to the browser over Server-Sent Events

- Added a relevance-threshold filter on retrieval so out-of-scope questions are refused instead of answered with fabricated context
- Identified and fixed a reflected XSS vulnerability in the chat interface by replacing unsafe HTML injection with safe DOM rendering
- Containerized the API and LLM services with Docker Compose and open-sourced the project on GitHub

Undergraduate Research Assistant — UAV Data & Vision Systems
Aug 2024 – Mar 2025

- Developed OpenCV-based computer vision pipelines for real-time object tracking in UAV simulations
- Processed geospatial routing data using OSMnx to improve autonomous navigation efficiency
- Simulated autonomous systems in Webots to evaluate navigation performance and system reliability
- Collaborated with research teams to improve tracking accuracy and system performance

Skills & Interests

- Languages: Python, JavaScript, TypeScript, C++, SQL
- Backend: Node.js, FastAPI, Flask, REST APIs
- Frontend: SvelteKit, HTML/CSS
- Cloud & DevOps: AWS (EC2, S3), Google Cloud Platform, Docker, CI/CD
- Databases: PostgreSQL, MongoDB, Firebase Firestore
- AI / Machine Learning: OpenCV, YOLOv8, FAISS, sentence-transformers, RAG, Ollama, Real-Time Data Processing
- Tools: Git, Linux

Certifications

- AWS Knowledge: Serverless
- Google Cloud: App Dev Environment
- Google Cloud: Load Balancing